

DATA 430 Technical Report Assignment 3: Decision Trees	Matthew Kolonich
Predicting Customer Segmentation Using Decision Tree Classification	
URL to dataset: https://www.kaggle.com/datasets/prathamtripathi/customersegmentation	

This template should be used in conjunction with the assignment instructions. The size of the text area below will expand to the length of your response; the area should not be interpreted as a required or suggested length of response. Responses within the text area should be single spaced with Times New Roman 12pt font. The body of the document will likely be 6-9 pages, not including the Appendix; length may vary depending on specifics of the analysis and the dataset. As needed, APA format in-text citations should be included, along with a full references list at the end of the document.

Overview
Problem Domain: give some background and context about the problem domain (application area). For instance, if you are doing the analysis for predicting heart disease, provide some context about the disease and include some interesting statistics about it. Also, discuss how the method is relevant for the chosen problem. Customer segmentation is one of the most important strategies businesses use to better understand their customer base and improve decision-making. Organizations collect large amounts of customer data. This includes demographic, financial, and behavioral information, to identify meaningful patterns among different groups of customers. These patterns can help businesses improve customer relationships, increase profitability, and create more targeted marketing strategies. In the telecommunications industry, customer segmentation plays an especially important role. This is because companies often offer multiple service plans designed for different customer needs. Some customers may prioritize affordability. Others may focus on premium services, data usage, or long-term service agreements. Understanding these differences allows organizations to better align their products and services with customer expectations. This can improve retention and revenue generation. The problem domain for this study focuses on predicting customer categories based on demographic and behavioral variables. The dataset includes variables such as region, tenure, age, income, marital status, education level, years of employment, retirement status, gender, and residential information. These variables provide useful insight into customer characteristics. They can also help explain why customers fall into different service categories. I find this problem domain particularly relevant because customer behavior has always interested me, especially through my years of experience in retail management. Working directly with customers has shown me that spending habits, income levels, and even life stages can strongly influence decision-making. While telecommunications is a different industry, many of the same patterns of customer behavior still apply. This makes customer segmentation

a practical and realistic machine learning problem to study.

Decision Tree Classification is highly relevant to this problem because it provides both predictive power and interpretability. Decision trees create branching decision paths that clearly show how customer attributes influence classification outcomes. According to Kubat (2017), one of the major strengths of decision trees is their ability to create understandable models while still identifying meaningful patterns in large datasets. This makes them especially valuable in business settings where understanding the reasoning behind a prediction can be just as important as the prediction itself.

Objective: clearly state the objective of the analysis in relation to the kind of algorithm you are employing. Use specific language as to what question(s) you are trying to answer using the specific analysis/modeling type.

The objective of this study is to develop and evaluate a Decision Tree Classification model to predict customer categories within a telecommunications dataset. The target variable for this analysis is custcat. Custcat classifies customers into four separate customer groups labeled A, B, C, and D. The independent variables include demographic and behavioral factors. This includes variables such as region, tenure, age, income, marital status, education level, employment years, retirement status, gender, and residence.

The primary goal of this analysis is not only to classify customers accurately, but also to better understand which customer characteristics have the greatest influence on segmentation. Decision Tree Classification is especially useful for this purpose because it creates an interpretable model that clearly shows how variables contribute to classification decisions. This allows businesses to better understand the reasoning behind the predictions.

From a business perspective, customer segmentation can provide important value. It can do this by helping organizations improve customer service, personalize marketing strategies, and increase customer retention. By helping identify patterns that distinguish one customer group from another, companies can make more informed decisions. This can effect decisions about pricing models, service offerings, and promotional campaigns. In the telecommunications industry, this can directly impact both customer satisfaction and long-term revenue.

A secondary objective of this study is to evaluate the effectiveness of Decision Trees as a machine learning tool for customer classification. This includes measuring the model's performance using classification metrics such as accuracy, precision, recall, and confusion

matrix analysis. These evaluation methods will help determine whether the model successfully meets the goal of identifying meaningful customer segments.

Personally, I find this objective important because understanding customer behavior has always been a major part of my professional experience in retail. Although this dataset focuses on telecommunications customers, the broader idea remains the same: the better a company understands its customers, the better it can serve them. This makes the analysis both practical and highly relevant to real-world business decision-making.

Analysis

Exploratory Analysis: describe the data including the source, the collection method, and variables. Perform exploratory analysis. Also, select few key variables (including the target variable for supervised learning) and study their distributions using plots such as histograms, box plot, bar chart, etc.

The customer segmentation dataset contains 1,000 observations and 12 variables. The dataset was designed to classify telecommunications customers into four customer categories labeled A, B, C, and D. The target variable, custcat, represents the customer category. The remaining eleven variables serve as predictors. These predictor variables include region, tenure, age, income, marital status, address, education level, years employed, retirement status, gender, and residential area.

The dataset was first examined to understand its structure and identify any potential preprocessing concerns. The review of the dataset showed that all 1,000 observations were complete. There were no missing values across any variables. This reduced the complexity of preprocessing. It allowed the model to use the full dataset without requiring imputation or removal of records.

The descriptive statistics provided several useful insights into customer behavior. The average customer age was approximately 41.68 years. The average income was approximately \$77.54. However, income showed substantial variability, with values ranging from \$9 to \$1,668. This large range suggests the presence of income outliers. This was confirmed through the income boxplot. While Decision Trees are generally robust to outliers compared to

other machine learning methods, identifying these extreme values remained important for understanding the data distribution.

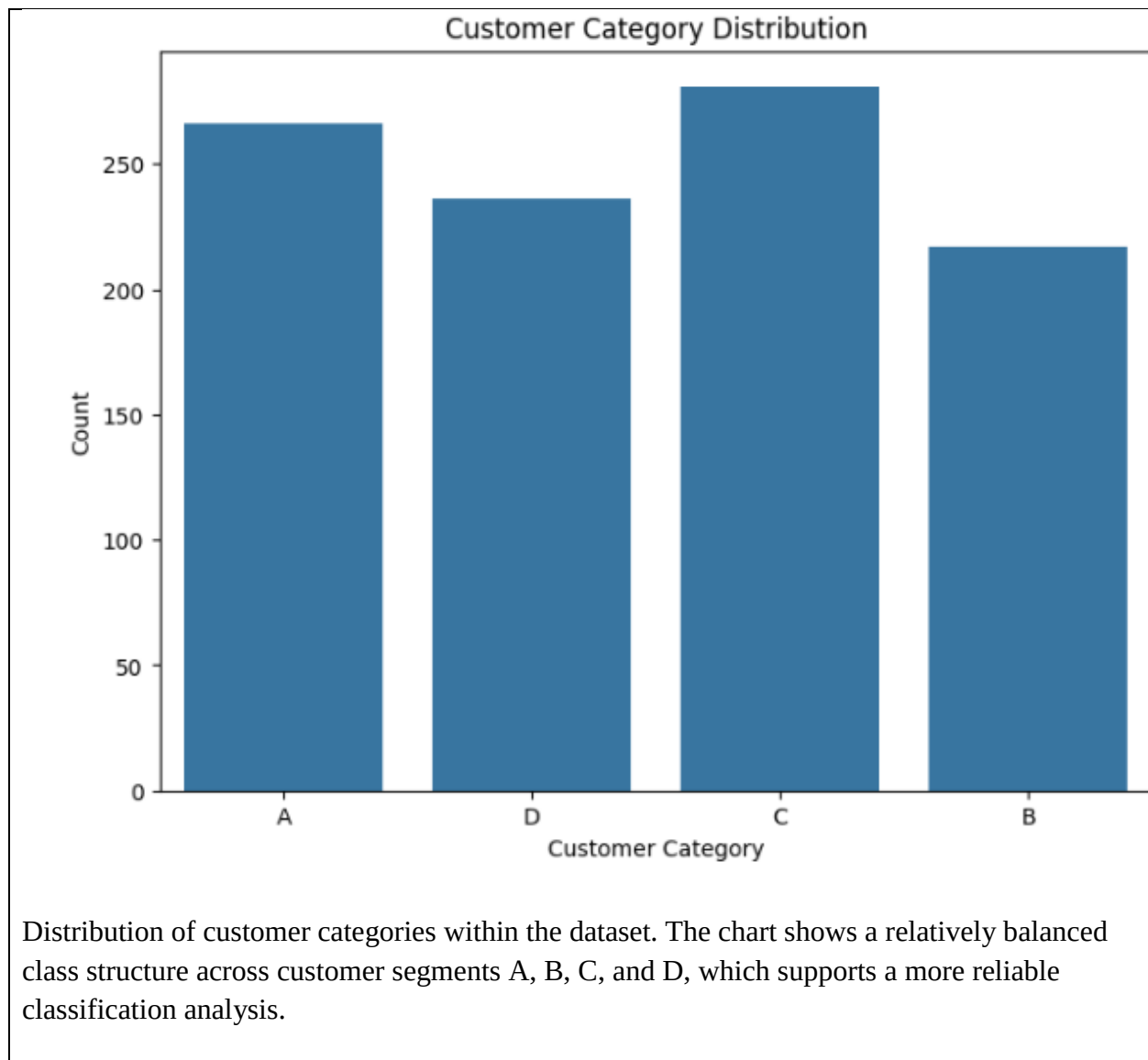
The target variable distribution showed that customer segmentation was relatively balanced across the four classes. Customer category C contained 281 observations, category A contained 266 observations, category D contained 236 observations, and category B contained 217 observations. This balanced structure created a better classification environment compared to previous assignments where severe class imbalance created additional modeling challenges. According to James et al. (2021), balanced target variables generally improve the reliability of classification model evaluation because predictions are less likely to be dominated by one majority class.

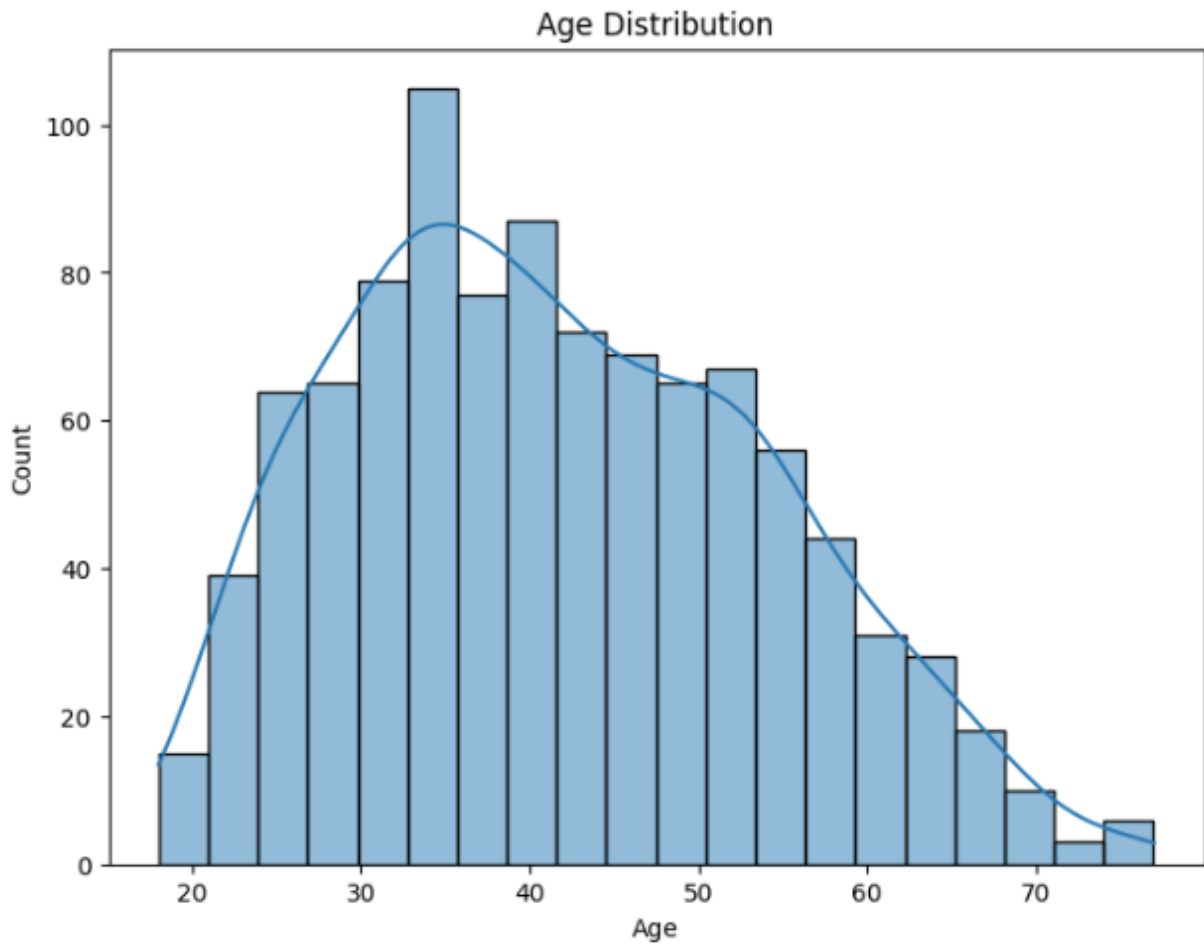
The correlation matrix revealed several meaningful relationships among predictor variables. The strongest positive relationship was observed between age and years employed ($r = 0.67$), followed closely by age and address duration ($r = 0.66$). Tenure and address duration also showed a moderate positive relationship ($r = 0.52$). These findings suggest that customer stability, age, and employment history may play important roles in determining customer segmentation. Kubat (2017) explains that understanding variable relationships before model construction can provide useful insight into potential predictive patterns and improve interpretation of model outcomes.

The visualizations further supported these observations. The customer category distribution chart confirmed the balanced structure of the target variable. The age distribution histogram showed that most customers fell between their early 30s and early 50s. This indicates that the dataset primarily captures middle-aged adults. The income distribution histogram showed a right-skewed distribution. Most customers are earning lower-to-middle income levels while a smaller number of customers represented much higher income levels. The boxplot clearly highlighted these high-income outliers.

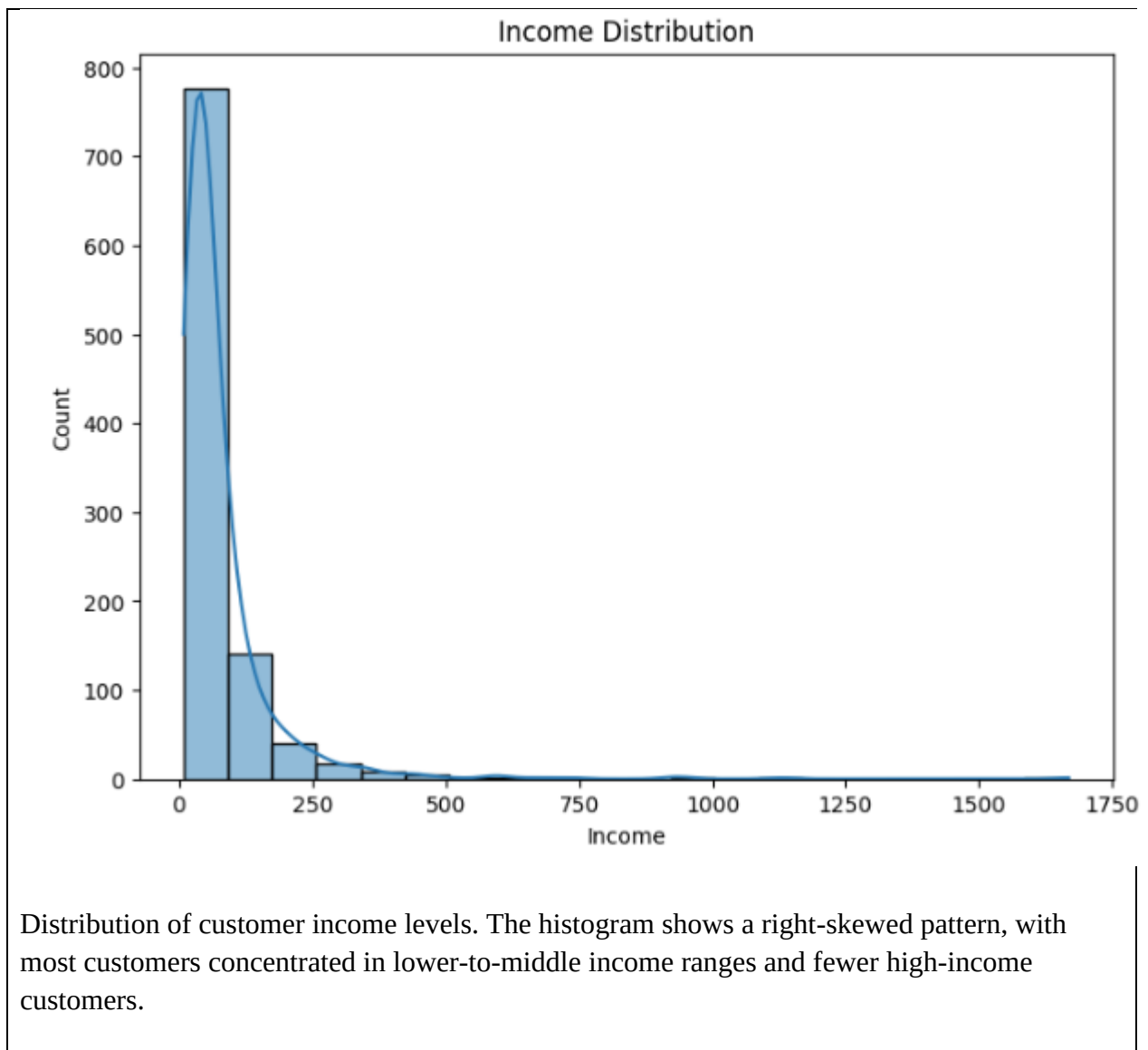
Personally, I found this part of the analysis especially interesting because it reinforces something I have seen throughout my retail career: customer behavior often follows patterns that businesses may overlook. Income, loyalty, and long-term stability often influence buying behavior, whether in telecommunications or retail. While the products differ, the patterns behind customer decision-making often remain surprisingly similar.

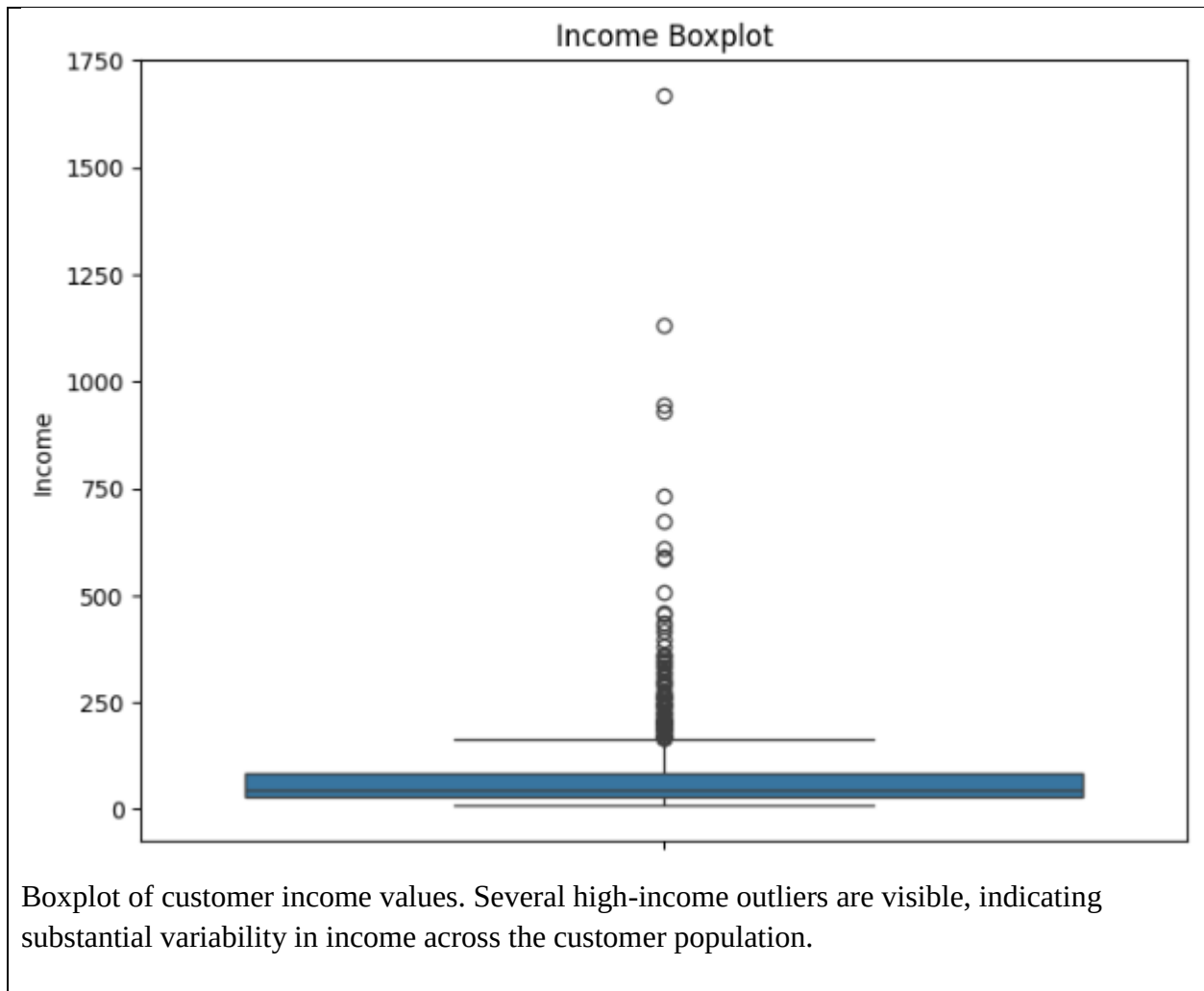
According to Bati (n.d.), exploratory analysis serves as an important foundation for machine learning because it allows researchers to better understand the data before applying predictive models. This step helped establish a clearer understanding of the customer population and prepared the dataset for the Decision Tree model used later in the analysis.

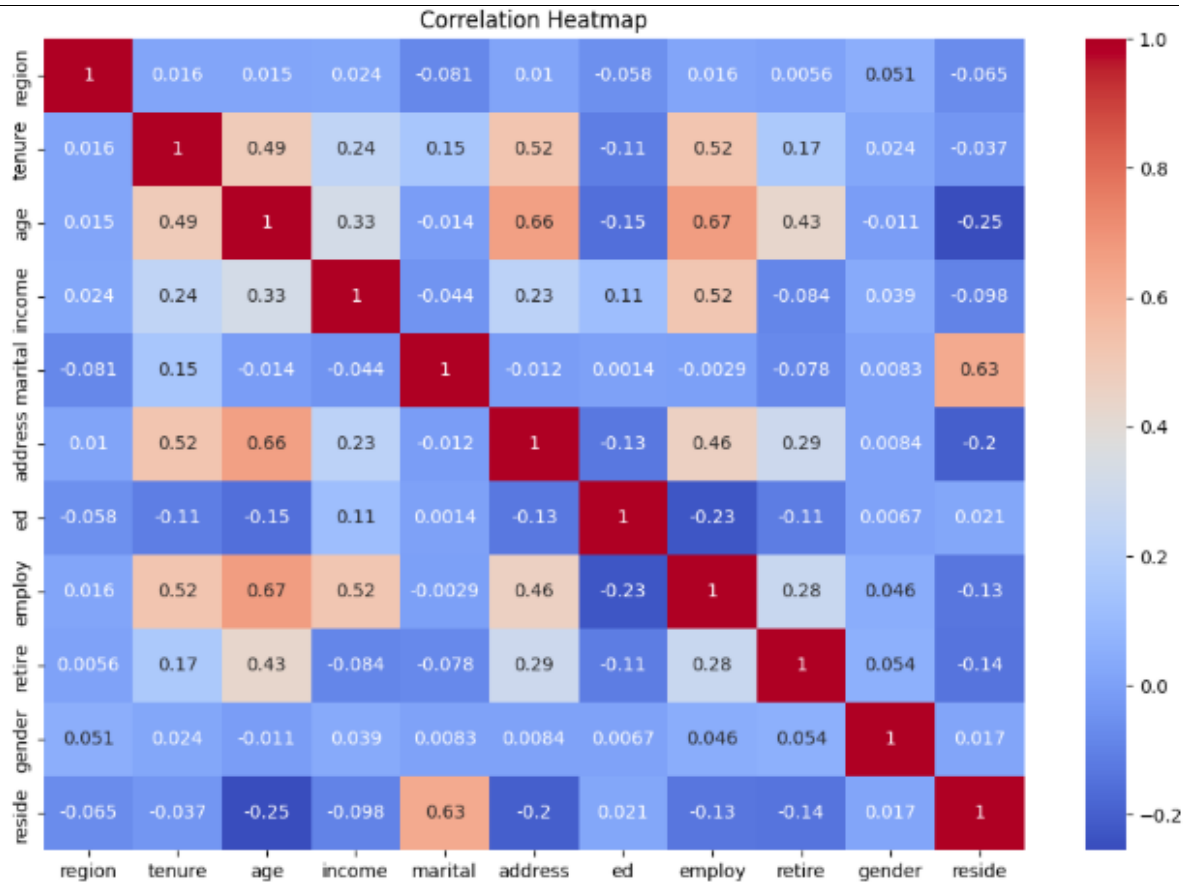




Distribution of customer ages in the dataset. Most customers fall between their early 30s and early 50s, indicating a concentration of middle-aged individuals within the sample.







Correlation heatmap of the predictor variables. The strongest positive relationships appear between age and employment years, and age and address duration, suggesting potential indicators of customer stability.

Preprocessing: armed with the exploratory analysis, perform the necessary preprocessing, both general and specific types appropriate for the modeling type being employed.

Before fitting the Decision Tree model, the dataset required several preprocessing steps to prepare it for classification analysis. The first step involved separating the independent variables from the dependent variable. In this study, the target variable was custcat,

representing customer category classifications A, B, C, and D. All remaining variables were used as predictors for the model.

Since the predictor variables were already stored as numerical values and contained no missing data, the preprocessing stage was relatively straightforward. No imputation techniques were necessary, and no categorical encoding was required for the independent variables. This reduced the complexity of data preparation compared to other machine learning projects where missing values or non-numeric variables can create additional challenges.

The target variable remained in its categorical format. This is because Decision Tree Classification can directly handle labeled class outputs without requiring transformation into numerical values. This preserved the interpretability of the customer categories during both model training and evaluation.

The dataset was then divided into training and testing sets using a 75% to 25% split. This resulted in 750 observations for model training and 250 observations for testing. A random state of 42 was used to ensure reproducibility of the results. In addition, stratified sampling was applied to preserve the original class distribution across both the training and testing datasets. According to James et al. (2021), maintaining class distribution during data splitting is important because it helps ensure that model evaluation remains representative of the full dataset.

Unlike methods such as logistic regression or K-nearest neighbors, Decision Trees do not require feature scaling or normalization. This is because the algorithm makes decisions based on split thresholds rather than distance calculations (Kubat, 2017). For this reason, no normalization or standardization was applied to the predictor variables.

An important observation during preprocessing was the presence of high-income outliers identified during exploratory analysis. Although these outliers were retained, this decision was intentional because Decision Trees are less sensitive to extreme values compared to linear models. Removing these values may have eliminated potentially meaningful customer behavior patterns.

Personally, I found this preprocessing stage much cleaner than previous projects. In many machine learning datasets, missing values and variable transformations often take a significant amount of time. In this case, the dataset structure allowed the focus to remain on model construction and interpretation rather than extensive data cleaning.

Model Fitting: explain the key steps and activities you perform to fit the model. Experiment (as appropriate) with parameters tuning. This is key, what separates highly accurate model from a less accurate ones is the amount of performance tuning performed.

The model fitting stage began by constructing an initial Decision Tree Classification model using the full training dataset. Decision Trees are widely used supervised learning algorithms. They recursively split data into smaller partitions based on decision rules designed to maximize class purity. According to Kubat (2017), Decision Trees are effective because they are highly interpretable and can model complex non-linear relationships while remaining relatively easy to understand.

The initial model was built using the default `DecisionTreeClassifier` settings in Python's Scikit-learn library. This first model served as a baseline for evaluating performance before hyperparameter tuning. After fitting the model on the training data, the baseline model achieved an accuracy of 28.0% on the testing dataset. While this performance was slightly better than random guessing across four classes(Barely), the confusion matrix and classification report showed weak performance across all customer categories. Precision, recall, and F1-scores remained low, indicating that the baseline model was not capturing strong segmentation patterns.

To improve performance, hyperparameter tuning was introduced using `GridSearchCV`. Grid search is a systematic method for evaluating multiple parameter combinations to identify the model configuration that produces the strongest performance. James et al. (2021) explain that hyperparameter tuning is essential in machine learning because model performance can vary significantly depending on parameter selection.

The first tuning process focused heavily on maximizing overall accuracy. This produced a model with a shallow tree depth of 2 and achieved a testing accuracy of 40.8%. This represented a significant improvement over the baseline model. Though, further analysis revealed an important limitation: the model completely failed to classify customer category B. This created a biased classification structure where one customer segment was effectively ignored. From a business perspective, this would reduce the usefulness of the segmentation model. All customer groups need to be represented.

Observing these limitations led to a second tuning process. It was performed with a broader hyperparameter search. The updated tuning process tested multiple values for splitting criteria, tree depth, minimum samples per split, minimum leaf size, and maximum feature selection. The final selected model used the following parameters:

- Criterion: Entropy
- Maximum Depth: 3
- Minimum Samples Split: 20
- Minimum Samples Leaf: 1
- Maximum Features: None

The final re-tuned model achieved an accuracy of 38.8%. Although this was slightly lower than the first tuned model, it produced a more balanced classification outcome. This is because it successfully predicted all four customer categories. This made the model more practical for customer segmentation analysis.

One important lesson during model fitting was recognizing that accuracy alone does not always represent the best model. Bati (n.d.) explains that classification performance should be evaluated across multiple metrics, especially in multi-class problems. In this case, the slightly lower accuracy model provided stronger overall segmentation coverage. It also better represented the real-world business problem.

This stage reminded me of real-world business decision-making. Sometimes the option that looks best on paper is not always the best in practice. In retail, I have seen similar situations where focusing too much on one metric can create blind spots elsewhere. Machine learning works the same way. The strongest model is often the one that balances performance with practical usefulness.

Results

Model Properties: explain the components of the fitted model and their characteristics. Leverage functions to summarize the model properties. Also, leverage visualization as required.

The final Decision Tree model selected after the second hyperparameter tuning process used the entropy criterion with a maximum tree depth of 3, a minimum sample split of 20, and a minimum leaf size of 1. The model was allowed to evaluate all predictor variables at each split. These parameter settings were selected through GridSearchCV after evaluating multiple combinations to improve class-level balance across the customer categories.

The final model produced a tree depth of 3 and contained 8 leaf nodes. This relatively shallow tree structure suggests that the model remained interpretable while still capturing meaningful customer segmentation patterns. According to Kubat (2017), one of the primary strengths of Decision Trees is their ability to provide transparent decision pathways, making it easier to understand how classifications are generated.

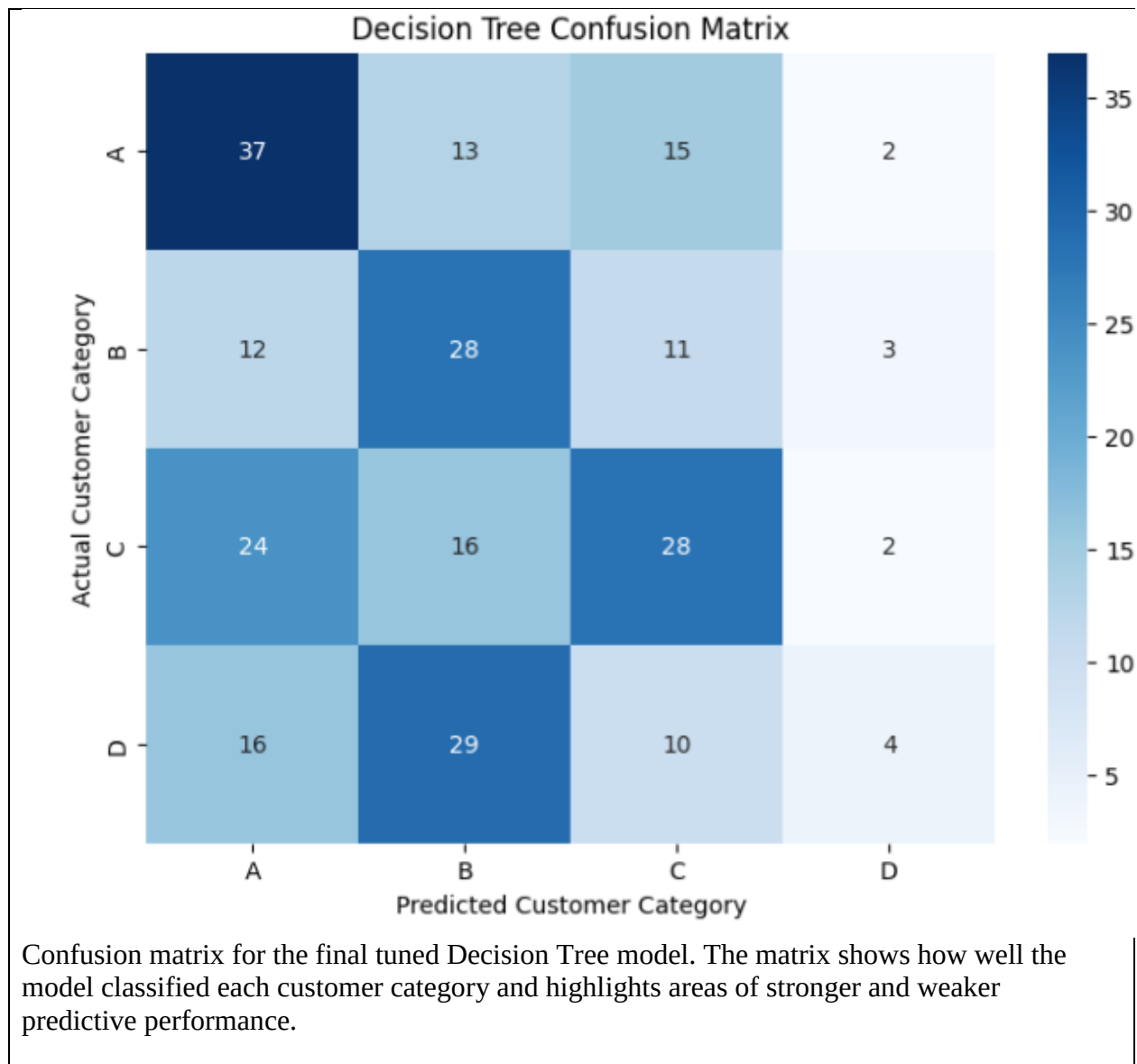
Feature importance analysis revealed that education level (ed) was the strongest predictor. It contributed approximately 49.7% of the total importance within the model. The second strongest predictor was tenure. It contributed approximately 38.3%. Address duration and residential area provided smaller contributions at 6.4% and 5.7% a piece. Interestingly,

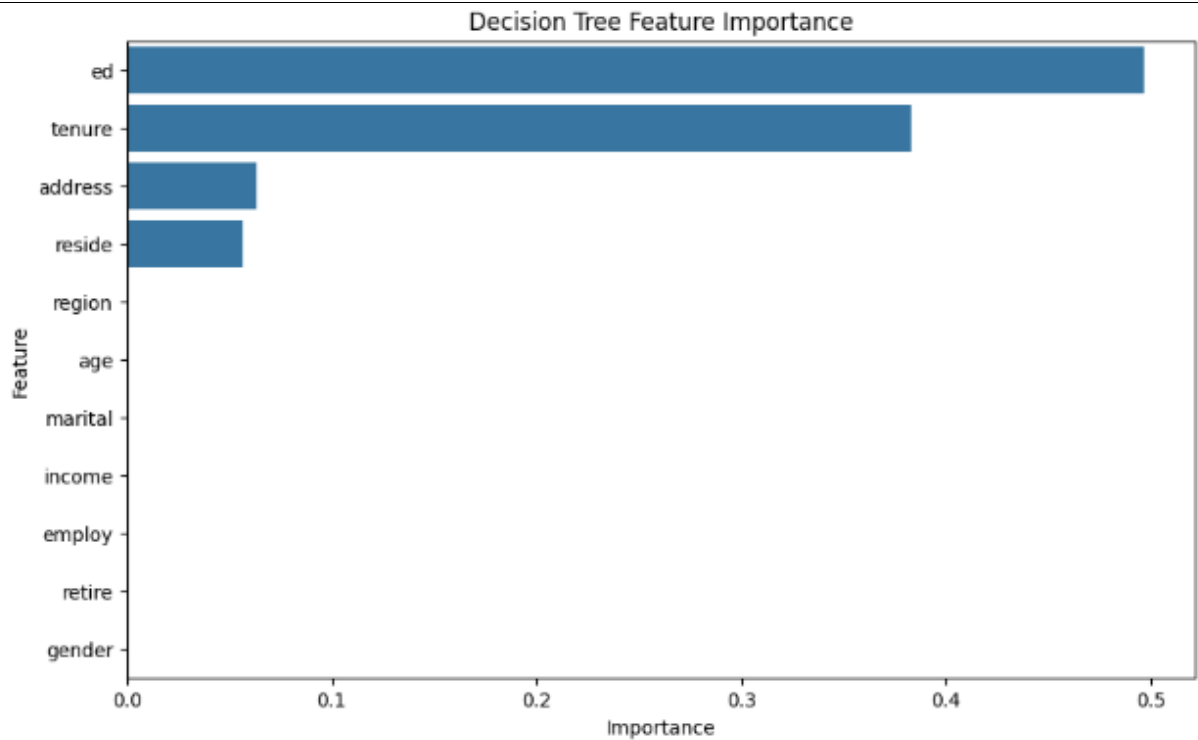
variables such as age, income, marital status, employment years, retirement status, and gender contributed little or no predictive value within the final model.

This result was somewhat surprising because variables such as income and age often appear important in customer behavior studies. However, James et al. (2021) explain that feature importance within tree-based models depends entirely on how much each variable reduces uncertainty during splitting. In this case, education and tenure provided the strongest classification separation.

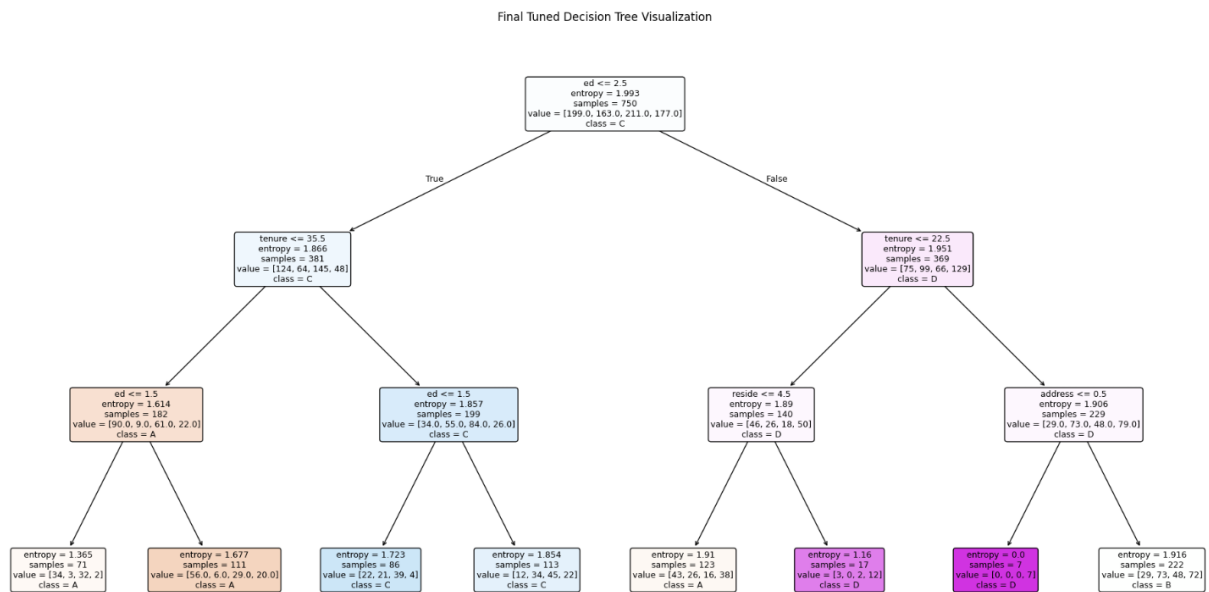
The confusion matrix showed how the final model distributed predictions across the four customer categories. Unlike the earlier tuned model, the final model successfully predicted all four classes. This was an important improvement because it created a more balanced segmentation model. This is even though the overall accuracy was slightly lower. From a structural perspective, this suggests the final model achieved broader class coverage. Which, in turn, improved its practical business usefulness.

I found the feature importance results interesting because they challenged some of my initial assumptions. I expected income or age to play a larger role. The model showed that customer tenure and education were much stronger predictors. This reinforces how machine learning can reveal patterns that may not be obvious through experience alone.





Feature importance rankings from the final Decision Tree model. Education and tenure contributed the most to customer classification, while several other variables had minimal impact.



Final tuned Decision Tree structure. The model illustrates how customer segmentation decisions were made based on key predictor variables and splitting criteria.

Output Interpretation: explain the result and interpret the final model output using terms that reflect the application area and in relation to the stated objective. This is where you check whether or not the stated objective is met.

The final re-tuned Decision Tree model achieved an overall testing accuracy of 38.8%. This represents an improvement of approximately 10.8% over the baseline model accuracy of 28.0%. While the overall accuracy remained moderate, the model provided meaningful insight into customer segmentation behavior and demonstrated measurable improvement through hyperparameter tuning.

The classification report revealed important differences in performance across the four customer categories. Customer category A produced the strongest balanced performance. Category A had a precision of 42%, recall of 55%, and an F1-score of 47%. This suggests that the model was most effective at identifying this customer segment. Customer category B also showed moderate performance, with a precision of 33%, recall of 52%, and an F1-score of 40%. This was especially important because the earlier tuned model failed to classify category B entirely.

Customer category C also performed reasonably well. The precision and recall values of category C were 44% and 40%, respectively. However, customer category D showed the weakest performance, with a recall of only 7% and an F1-score of 11%. This indicates that the model struggled to identify category D correctly. It frequently misclassified these customers into other categories.

The confusion matrix supports this interpretation by showing that category D customers were often predicted as category B or category A. This suggests overlapping feature patterns between those customer groups. This makes separation more difficult. According to James et al. (2021), overlapping feature distributions often reduce classification performance because the model has difficulty identifying clear decision boundaries.

One of the most important findings during this analysis was the tradeoff between overall accuracy and class coverage. The first tuned model achieved slightly higher accuracy at 40.8%, but it completely ignored customer category B. The re-tuned model lowered overall accuracy slightly to 38.8% but successfully predicted all four customer classes. This created a more balanced and realistic segmentation model.

This result reinforced an important machine learning principle discussed by Bati (n.d.): performance metrics must be interpreted in context. Accuracy alone does not always indicate the strongest model. This is especially true in multi-class classification problems. Precision, recall, and F1-score often provide a more complete picture of model effectiveness.

This part of the project was especially valuable because it reflects something I have seen in business for years. Numbers alone do not always tell the full story. A higher number may look better at first, but if important customer groups are ignored, the business decision may actually be weaker. This project helped reinforce how machine learning evaluation requires looking deeper than a single performance metric.

Evaluation: employ appropriate metrics to quantitatively evaluate the performance of the fitted model. For supervised classification, this includes simple accuracy, precision & recall (or sensitivity & specificity), all of which can be generated from a confusion matrix, or ROC.

The final Decision Tree model demonstrated moderate success in predicting customer segmentation categories. With an overall testing accuracy of 38.8%, the model performed noticeably better than the baseline model. The baseline achieved only 28.0% accuracy. While the improvement was meaningful, the overall accuracy still indicates that customer segmentation remains a complex classification problem with overlapping customer characteristics.

One of the strongest aspects of the final model was its ability to classify all four customer categories. This represented an important improvement over the earlier tuned model. The first model completely failed to classify category B despite producing slightly higher overall accuracy. This highlights an important machine learning principle: model effectiveness should not be measured by accuracy alone. According to James et al. (2021), evaluating classification models requires balancing multiple performance metrics, including precision, recall, and F1-score, to better understand model behavior across all classes.

The final model performed strongest in customer category A. This is where recall reached 55%. Category B, was also strong, where recall reached 52%. These results suggest that the model was effective at identifying certain customer groups based on education, tenure, and residential stability. However, the model performed poorly for customer category D. Here recall dropped to only 7%. This indicates that category D likely shares similar characteristics with other customer groups, making separation more difficult.

From a business perspective, the model still provides useful insights despite its limitations. The feature importance analysis showed that education and tenure were the most influential variables in determining customer category. This information could help telecommunications companies better understand which customer characteristics may drive long-term service preferences or retention behaviors. Kubat (2017) explains that one of the major strengths of

Decision Trees is that they not only make predictions but also reveal which features matter most in the decision-making process.

Another important takeaway from this evaluation was the tradeoff between model simplicity and predictive performance. The final model remained relatively shallow with a tree depth of only 3. This improved interpretability but may have limited its ability to capture more complex relationships. Bati (n.d.) notes that simpler models are often easier to interpret, but they may sacrifice predictive power when dealing with complex multi-class problems.

I believe the model was successful because it achieved the main objective of improving customer segmentation understanding rather than simply maximizing accuracy. In business, even moderate predictive improvements can provide value when they lead to better decision-making. This project reinforced that machine learning is not always about building perfect models, but about building useful ones.

Conclusion

Summary: highlight the main findings in relation to the stated objective. You don't need to discuss the details of the analysis and the model such as accuracy here, just focus on the key findings.

The purpose of this study was to develop and evaluate a Decision Tree Classification model for customer segmentation within a telecommunications dataset. The model used demographic and behavioral variables. These included tenure, age, income, education, marital status, and residential information. The variables were used to predict customer category classifications across four customer groups. The primary objective was not only to build a predictive model, but also to better understand which customer characteristics most strongly influence segmentation outcomes.

The analysis began with exploratory data analysis. This revealed a balanced target variable distribution and several meaningful relationships among the predictor variables. The dataset contained no missing values, which allowed the preprocessing stage to remain relatively simple. One notable observation was the presence of significant income outliers. Although, these values were retained because Decision Trees are generally less sensitive to extreme values than other machine learning models.

The model fitting process demonstrated the importance of iterative tuning. The baseline Decision Tree model achieved an accuracy of 28.0%, which showed limited predictive ability. The first tuning process improved accuracy to 40.8%, but failed to classify one of the customer categories entirely. A second tuning process was then performed to improve class-level balance. The final model achieved 38.8% accuracy while successfully predicting all four customer categories. This reinforced the idea that higher accuracy does not always mean a better model.

Feature importance analysis revealed that education level and customer tenure were the most influential predictors in customer segmentation. This finding was particularly valuable because it showed that customer stability and educational background had greater predictive influence than variables such as age or income. According to Kubat (2017), one of the strengths of Decision Trees is their ability to reveal these important relationships while maintaining model transparency.

One of the most important lessons learned from this project was the importance of evaluating multiple performance metrics rather than relying only on accuracy. James et al. (2021) explain that precision, recall, and F1-score often provide a more complete understanding of model performance, especially in multi-class classification problems. This project clearly demonstrated that principle. It showed the model with slightly lower accuracy ultimately produced stronger and more balanced customer classification results.

I found this project helped strengthen my understanding of how machine learning can support business decision-making. Through my experience in retail, I have often seen customer behavior patterns develop over time. This project showed how machine learning can quantify those patterns in ways that are not always obvious through observation alone. It reinforced the idea that machine learning is not simply about prediction, but about understanding behavior in ways that can improve business strategy.

Limitations & Improvement areas: discuss the limitations of the analysis and identify potential improvement areas for future work. This could be related to the data, algorithm, or a combination of the two.

Although the final Decision Tree model provided useful insights into customer segmentation, several limitations were identified during the analysis. The most significant limitation was the overall predictive performance. While the final model improved upon the baseline model and achieved an accuracy of 38.8%, this still indicates that a large portion of

customer classifications were incorrect. This suggests that customer segmentation within this dataset is a complex problem with overlapping customer characteristics that are difficult to separate clearly.

A major limitation was the model's uneven performance across customer categories. While customer categories A, B, and C showed moderate predictive success, customer category D performed poorly. D had a recall of only 7%. This indicates that the model struggled to identify that specific customer group. It frequently misclassified category D into other categories. According to James et al. (2021), this type of class confusion often occurs when feature distributions overlap significantly, reducing the model's ability to establish clear decision boundaries.

Another limitation was the relatively shallow depth of the final Decision Tree. Although a depth of 3 improved interpretability, it may have limited the model's ability to capture more complex customer relationships. Deeper trees may improve predictive performance but they also increase the risk of overfitting. Kubat (2017) explains that balancing model complexity and generalization remains one of the most important challenges in tree-based learning.

The feature importance analysis also revealed that several variables contributed little or no predictive value. These included age, income, gender, retirement status, and employment years. While these variables may still have business significance, they were not useful within the current model structure. This suggests that either the variables themselves lack predictive power or that the relationships may be more complex than a single Decision Tree can capture.

There are several ways this analysis could be improved in future studies. One improvement would be testing more advanced tree ensemble methods such as Random Forest or Gradient Boosting. These methods often improve classification performance by combining multiple trees and reducing variance (Bati, n.d.). Another improvement would be feature engineering. This is where new variables or interaction terms could be created to capture more complex customer behavior patterns.

Additionally, clustering methods in the next phase of this project may provide deeper insights into customer segmentation. Clustering can help by identifying hidden groupings without relying on labeled target variables. This may reveal customer relationships that supervised classification methods cannot fully capture.

In conclusion, one of the biggest lessons I learned from this project is that machine learning models often expose complexity that is not immediately visible. In business, customer behavior can appear straightforward on the surface, but the deeper patterns are often far more complicated. This project reinforced the importance of patience, iteration, and critical evaluation when building predictive models.

Appendix

First 10 Rows:

\	region	tenure	age	income	marital	address	ed	employ	retire	gender
0	2	13	44	64	1	9	4	5	0	0
1	3	11	33	136	1	7	5	5	0	0
2	3	68	52	116	1	24	1	29	0	1
3	2	33	33	33	0	12	2	0	0	1
4	2	23	30	30	1	9	1	2	0	0
5	2	41	39	78	0	17	2	16	0	1
6	3	45	22	19	1	2	2	4	0	1
7	2	38	35	76	0	5	2	10	0	0
8	3	45	59	166	1	7	4	31	0	0
9	1	68	41	72	1	21	1	22	0	0

```
reside custcat
```

0	2	A
1	6	D
2	2	C
3	1	A
4	4	C
5	1	C
6	5	B
7	3	D
8	5	C
9	3	B

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

Data columns (total 12 columns):

#	Column	Non-Null	Count	Dtype
0	region	1000	non-null	int64
1	tenure	1000	non-null	int64
2	age	1000	non-null	int64
3	income	1000	non-null	int64
4	marital	1000	non-null	int64
5	address	1000	non-null	int64
6	ed	1000	non-null	int64
7	employ	1000	non-null	int64
8	retire	1000	non-null	int64
9	gender	1000	non-null	int64
10	reside	1000	non-null	int64
11	custcat	1000	non-null	object

```
dtypes: int64(11), object(1)
```

```
memory usage: 93.9+ KB
```

None

Missing Values:

```
region      0
tenure      0
age         0
income      0
marital     0
address     0
ed          0
employ      0
retire      0
gender      0
reside      0
custcat     0
dtype: int64
```

Summary Statistics:

	region	tenure	age	income	marital \
count	1000.0000	1000.000000	1000.000000	1000.000000	1000.000000
mean	2.0220	35.526000	41.684000	77.535000	0.495000
std	0.8162	21.359812	12.558816	107.044165	0.500225
min	1.0000	1.000000	18.000000	9.000000	0.000000
25%	1.0000	17.000000	32.000000	29.000000	0.000000
50%	2.0000	34.000000	40.000000	47.000000	0.000000
75%	3.0000	54.000000	51.000000	83.000000	1.000000
max	3.0000	72.000000	77.000000	1668.000000	1.000000

	address	ed	employ	retire	gender \
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	11.551000	2.671000	10.987000	0.047000	0.517000
std	10.086681	1.222397	10.082087	0.211745	0.499961
min	0.000000	1.000000	0.000000	0.000000	0.000000
25%	3.000000	2.000000	3.000000	0.000000	0.000000
50%	9.000000	3.000000	8.000000	0.000000	1.000000
75%	18.000000	4.000000	17.000000	0.000000	1.000000
max	55.000000	5.000000	47.000000	1.000000	1.000000

	reside
count	1000.000000
mean	2.331000
std	1.435793
min	1.000000
25%	1.000000
50%	2.000000
75%	3.000000
max	8.000000

Customer Category Distribution:

```
custcat
C      281
A      266
D      236
B      217
```

Name: count, dtype: int64

Correlation Matrix:

	region	tenure	age	income	marital	address
ed \						
region	1.000000	0.016388	0.015034	0.023765	-0.080637	0.010320 -
0.057952						
tenure	0.016388	1.000000	0.490217	0.242912	0.154171	0.522952 -
0.109797						
age	0.015034	0.490217	1.000000	0.327948	-0.014433	0.660435 -
0.154857						
income	0.023765	0.242912	0.327948	1.000000	-0.043984	0.226781
0.113939						
marital	-0.080637	0.154171	-0.014433	-0.043984	1.000000	-0.012051
0.001400						
address	0.010320	0.522952	0.660435	0.226781	-0.012051	1.000000 -
0.128330						
ed	-0.057952	-0.109797	-0.154857	0.113939	0.001400	-0.128330
1.000000						
employ	0.016335	0.520288	0.670047	0.516349	-0.002891	0.463291 -
0.228580						
retire	0.005595	0.171143	0.429440	-0.083651	-0.078109	0.292503 -
0.114229						
gender	0.050596	0.023533	-0.011419	0.038706	0.008345	0.008363
0.006704						
reside	-0.065158	-0.036592	-0.254993	-0.097591	0.625999	-0.201645
0.021045						

	employ	retire	gender	reside
region	0.016335	0.005595	0.050596	-0.065158
tenure	0.520288	0.171143	0.023533	-0.036592
age	0.670047	0.429440	-0.011419	-0.254993
income	0.516349	-0.083651	0.038706	-0.097591
marital	-0.002891	-0.078109	0.008345	0.625999
address	0.463291	0.292503	0.008363	-0.201645
ed	-0.228580	-0.114229	0.006704	0.021045
employ	1.000000	0.277401	0.046016	-0.134683
retire	0.277401	1.000000	0.053906	-0.140120
gender	0.046016	0.053906	1.000000	0.016556
reside	-0.134683	-0.140120	0.016556	1.000000

Independent Variables:

```
Index(['region', 'tenure', 'age', 'income', 'marital', 'address', 'ed',
      'employ', 'retire', 'gender', 'reside'],
      dtype='object')
```

Dependent Variable:

```
custcat
```

Training Set Shape:

```
(750, 11)
```

Testing Set Shape:

```
(250, 11)
```

Initial Decision Tree Accuracy:
0.28

Initial Decision Tree Confusion Matrix:

```
[[21 14 17 15]
 [14 11 14 15]
 [15 17 23 15]
 [11 19 14 15]]
```

Initial Decision Tree Classification Report:

	precision	recall	f1-score	support
A	0.34	0.31	0.33	67
B	0.18	0.20	0.19	54
C	0.34	0.33	0.33	70
D	0.25	0.25	0.25	59
accuracy			0.28	250
macro avg	0.28	0.27	0.28	250
weighted avg	0.28	0.28	0.28	250

Best Parameters from GridSearchCV:

```
{'criterion': 'entropy', 'max_depth': 3, 'max_features': None,
 'min_samples_leaf': 1, 'min_samples_split': 20}
```

Best Cross-Validation Accuracy:

0.3933333333333333

Final Tuned Decision Tree Accuracy:

0.388

Final Tuned Decision Tree Confusion Matrix:

```
[[37 13 15  2]
 [12 28 11  3]
 [24 16 28  2]
 [16 29 10  4]]
```

Final Tuned Decision Tree Classification Report:

	precision	recall	f1-score	support
A	0.42	0.55	0.47	67
B	0.33	0.52	0.40	54
C	0.44	0.40	0.42	70
D	0.36	0.07	0.11	59
accuracy			0.39	250
macro avg	0.39	0.38	0.35	250
weighted avg	0.39	0.39	0.36	250

Feature Importance:

	Feature	Importance
6	ed	0.496616
1	tenure	0.383084

5	address	0.063542
10	reside	0.056758
0	region	0.000000
2	age	0.000000
4	marital	0.000000
3	income	0.000000
7	employ	0.000000
8	retire	0.000000
9	gender	0.000000

Decision Tree Model Properties:

Tree Depth: 3

Number of Leaves: 8

Criterion: entropy

Classes: ['A' 'B' 'C' 'D']

Model Comparison:

Initial Accuracy: 0.28

Final Tuned Accuracy: 0.388

Accuracy Improvement: 0.10799999999999998

References

- Bati, H. (n.d.). *Decision trees and tree ensemble methods* [Course lecture]. University of Maryland Global Campus.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
- Kubat, M. (2017). *An introduction to machine learning* (2nd ed.). Springer. <https://doi.org/10.1007/978-3-319-63913-0>
- Han, J., Pei, J., & Kamber, M. (2011). *Data mining: Concepts and techniques* (3rd ed.). Morgan Kaufmann.
- Tan, P. N., Steinbach, M., & Kumar, V. (2019). *Introduction to data mining* (2nd ed.). Pearson.